

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of M.Sc. (Mathematics) Examination March 2018
MA711 Complex Analysis

Date: 26/03/2018, Monday Time: 10:00 a.m. to 10:45 a.m. Maximum Marks: 20

MCQ

Important Instructions:

1. Tick the correct answer and it should be written in question paper itself.
 2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 3. Each question carries one mark.
 4. All notations are standard.
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Q - I Choose the correct answer from the given options for the following questions.

1. $\text{Log}(-1) = \underline{\hspace{2cm}}$.

- (a) πi (b) ∞ (c) $2\pi i$ (d) 0

2. The function $f(z) = u(x, y) + iv(x, y)$ is differentiable at $z_0 = x_0 + iy_0$ then $f'(z_0) = \underline{\hspace{2cm}}$.

- (a) $u_x(x_0, y_0) + v_x(x_0, y_0)$ (c) $u_y(x_0, y_0) + v_x(x_0, y_0)$
(b) $u_x(x_0, y_0) - v_x(x_0, y_0)$ (d) None of these

3. $\text{Arg}(-1 - i) = \underline{\hspace{2cm}}$.

- (a) $\frac{1}{4}\pi$ (b) $-\frac{1}{4}\pi$ (c) $-\frac{5}{4}\pi$ (d) $-\frac{3}{4}\pi$

4. The function $u(x, y)$ is harmonic if _____.
 (a) $u_x + u_y = 0$ (b) $u_{xy} + u_{xy} = 0$ (c) None of these (d) $u_{xx} + u_{yy} = 0$
5. The radius of convergence of $\sum_{n=0}^{\infty} (4n+1)z^n$ is _____.
 (a) 4 (b) ∞ (c) 0 (d) 1
6. For any $z \in \mathbb{C}$, $|e^z| =$ _____.
 (a) e^x (b) 1 (c) e^y (d) $2\pi i$
7. If $z_1, z_2 \in \mathbb{C}$ with $z_2 \neq 0$ then $\log\left(\frac{z_1}{z_2}\right) =$ _____.
 (a) $\log z_1 + \log z_2$ (b) $\log z_1 \log z_2$ (c) $\log z_1 - \log z_2$ (d) $\log z_2 / \log z_1$
8. The isolated singular point(s) of the function $\frac{1}{\sin(\pi/z)}$ is(are) _____.
 (a) $n\pi, n \in \mathbb{N}$ (c) 0
 (b) $\frac{1}{n}, n \in \mathbb{N}$ (d) $(2n+1)\frac{\pi}{2}, n \in \mathbb{N}$
9. The Cauchy-Riemann equations are given by _____.
 (a) $u_x = v_y, u_y = v_x$ (c) $u_x = v_y, u_y = v_y$
 (b) $u_x = v_x, u_y = v_x$ (d) $u_x = v_y, u_y = -v_x$
10. If C denotes the positively oriented circle $|z - z_0| = R$, then the value of $\int_C (z - z_0)^2 dz$ is _____.
 (a) 0 (b) $2\pi i$ (c) 2π (d) $\frac{1}{2\pi i}$
11. A non-constant entire function is _____.
 (a) discontinuous (c) unbounded
 (b) bounded (d) None of these
12. The impact of the function $f(z) = iz$ on the complex plane is _____.
 (a) translate z by 1 unit
 (b) reflects z about the line
 (c) rotates z by the right angle counter-clockwise
 (d) rotates z by the right angle clockwise

13. The function $f(z) = \frac{1}{\cos z}$ _____.

- (a) has not a pole at $z = \frac{\pi}{2}$ (c) is an entire function
(b) is analytic at $z = \frac{\pi}{2}$ (d) None of these

14. If $f(z)$ is analytic within and on a simple closed curve C and z_0 is any point within C , then _____.

- (a) $\int_C \frac{f(z)}{(z-z_0)} dz = 0$ (c) $\int_C \frac{f(z)}{(z-z_0)} dz = 2\pi i f(z_0)$
(b) $\int_C \frac{f(z)}{(z-z_0)} dz = 2\pi i$ (d) $\int_C \frac{f(z)}{(z-z_0)} dz = 2\pi i f(0)$

15. The zeros of $f(z) = \cos z$ are _____.

- (a) $\frac{(2k+1)\pi}{2}, \quad k \in \mathbb{Z}$ (c) $k\pi i, \quad k \in \mathbb{Z}$
(b) $k\pi, \quad k \in \mathbb{Z}$ (d) $k\pi + i, \quad k \in \mathbb{Z}$

16. The Möbius transformation $\frac{az+b}{cz+d}, a, b, c, d \in \mathbb{C}$ is defined if _____.

- (a) $ad - bc < 0$ (b) $ad - bc > 0$ (c) $ad - bc \neq 0$ (d) $ad - bc = 0$

17. Let function $f(z)$ be an analytic function in domain D . Then f is conformal if _____.

- (a) $f'(z) = 0$ (b) $f'(z) \neq 0$ (c) $f'(z) = 0$ (d) $f'(z) \neq 0$

18. The Maclaurin series $\sum_{k=0}^{\infty} \frac{z^k}{k!}$ of e^z converges for _____.

- (a) for $|z| < 1$ (c) for all $z \in \mathbb{C}$
(b) for $|z| \leq 1$ (d) None of these

19. The period of $f(z) = e^z$ is _____.

- (a) π (b) $2\pi i$ (c) 2π (d) π

20. The function $f(z) = z^{2018} + z^{2017} + \dots + z + 1$ _____.

- (a) has at least one zero
(b) has not a non-zero derivative of order 2018
(c) does not satisfy C-R equations
(d) None of these

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MA711 Complex Analysis

Date: 26/03/2018, Monday Time: 10:45 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

1. **Section I and Section II must be written in separate answer books..**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations are standard.

SECTION I

Q - II (A) Attempt any TWO of the following. [8]

- (1) Show that $u(x, y)$ is harmonic in some domain and find a harmonic conjugate $v(x, y)$ if $u(x, y) = 2x(1 - y)$.
- (2) If C_R denote the upper half circle $|z| = R$ ($R > 2$) taken in counterclockwise direction. Then show that

$$\left| \int_{C_R} \frac{2z^2 - 1}{z^4 + 5z^2 + 4} dz \right| \leq \frac{\pi R (2R^2 + 1)}{(R^2 - 1)(R^2 - 4)}.$$

- (3) State and prove Morera's theorem.
- (4) Find the Maclaurin's series expansion of $7z^3 e^{\frac{1}{z}}$, $z \neq 0$.

(B) Attempt any SIX of the following. [12]

- (1) For $z, w \in \mathbb{C}$, prove that $|z + w| \leq |z| + |w|$.
- (2) Find $\text{Arg}(i + 1)$.
- (3) Check whether the function $f(z) = |z|$ is analytic or not.
- (4) State Liouville's theorem.
- (5) Find the fixed points of $w = \frac{z - 1}{z + 1}$.
- (6) State maximum modulus principle.
- (7) State fundamental theorem of algebra.
- (8) Evaluate $\int_C e^{3z} \cos z dz$, where C is the contour with vertices $z = 1, z = i, z = 2 + 3i$.
- (9) For any $z \in \mathbb{C}$, Prove that $\sin(iz) = i \sinh(z)$.

SECTION II

Q - III (A) Attempt any THREE of the following. [18]

- (1) State and prove the Cauchy's Theorem.
- (2) (i) If C denotes the positively oriented boundary of the square whose vertices are $(\pm 2, \pm 2)$ then evaluate $\int_C \frac{e^{-z} dz}{z - (\pi/2i)}$.
(ii) Find the residue of $f(z) = \frac{1}{z + z^2}$ at $z = 0$.
- (3) Evaluate the integral $\int_{|z|=3} \frac{z+1}{z^2-2z} dz$ using residue theorem.
- (4) (i) Prove that the Möbius transformation is one-one function.
(ii) Find a bilinear transformation which maps the points $-i, 0, i$ of the z -plane onto the points $-1, i, 1$ of the w -plane respectively.
- (5) If a function f is analytic throughout an open disk $|z - z_0| < R_0$ then prove that for each z in this disk, $f(z)$ has a series representation
$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \text{ where } a_n = \frac{f^{(n)}(z_0)}{n!}, \quad n = 0, 1, 2, \dots$$

(B) Attempt any TWO of the following. [12]

- (1) State and prove Cauchy integral formula.
 - (2) Show that if $c_1 < 0$, then the image of the half plane $x < c_1$ under the transformation $w = \frac{1}{z}$ is an interior of the circle. What is the image when $c_1 = 0$ under the same transformation?
 - (3) Expand the function $f(z) = \frac{z+4}{(z+3)(z-1)^2}$, in Laurent's series expansion valid in (i) $0 < |z-1| < 4$ (ii) $|z-1| > 4$.
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